

046211 – Deep Learning – Project Guidelines and Grade Structure

The goal of the project is to let you gain hands-on experience with one of the many challenges that may be solved with machine learning and deep learning in particular. In the process, you will learn many new things that you may have not seen in the course, but the tools we provided you with will definitely help you face the challenge. We hope you will be able to use this project in the future in the form of a subject in job interviews, attracting recruiters on LinkedIn and maybe even find a good topic to research in the future (maybe graduate studies?).

Previous Projects: <https://taldatech.github.io/ee046211-deep-learning/>

Personal Meetings: towards the end of the semester, the instructors will hold 5/10-minute personal meetings (Zoom) with the pairs of students to help them with any issues they face (WE GOT YOU). Feel free to use the forum at any time for any assistance (you can also set-up office hours).

Ethics Statement: Project Stakeholders and Implications Assessment

Please assess the potential impact of your project on various stakeholders. Consider who the stakeholders ("persons of interest") are in your project and analyze the implications of your project for or on them. This assessment will help you understand the ethical considerations involved and ensure that your project is responsible and beneficial to society. **The ethics statement should follow the template provided below.** Please include the ethics statement in your **reports** (you can optionally include it in your presentations as well, within the presentation time-limit). See instructions and examples for this statement below. Please note that the ethics assignment is graded by ethics professionals from the humanities faculty.

Grade Structure: This project is 30% of your final grade. The project will be done in **pairs**. The grade in the project will be based upon:

- 50% Presentation in class
 - In the final week of the semester, you will present the project and its results in class.
 - You will need to prepare up to a **6-minute presentation** (~10 slides PowerPoint presentation) which **you will record** on video (Zoom/OBS Studio and etc...). Please do not exceed 6 minutes.
 - After the presentation, the staff will ask the students a couple of questions.
- 25% Detailed report with code files
 - You will need to submit a report/documentation of your projects.
 - Up to 8 pages.
 - You are free design the structure you report but it should include: introduction (project goal, motivation, previous work), method (explanation of the algorithm, of the architecture, loss function and etc.), experiments and results (explanation of the dataset, the experiments you performed to validate your method and their results compared to previous method/s) and a conclusion/future work.
 - Please add references to previous papers and works.
 - We highly recommend working in Overleaf (register with your campus emails to get premium accounts so you can work together on the same file) but you can also use Word/LyX...
 - 5 points will be assigned for the ethics statement.
- 25% Well-organized GitHub repository
 - **IMPORTANT** – this is how the world will see your project, so we are giving this a large weight in the final grade. We expect a well-organized, well-structured GitHub

repository of your code. This should include your documented code (USE COMMENTS WHEN YOU CODE) and a VERY detailed README file of how to run your code and how the repository is organized.

- Examples: you can copy the README file of repositories you like and modify it to your projects.
- Here are links to some of Tal Daniel's projects (<https://github.com/taldatech>), feel free to copy the README file: <https://github.com/taldatech/ee046211-deep-learning> , <https://github.com/taldatech/pytorch-ls-dqn> , <https://github.com/taldatech/image2pencil-drawing>

Good luck!

Ethics Statement Instructions:

Following are general instructions, and at the end you will find the template that you should fill in your report (YOU MUST USE THE TEMPLATE).

Identify Stakeholders:

List all the stakeholders who may be affected directly or indirectly by your project. This could include end-users, communities, organizations, policymakers, and other relevant parties.

Analyze Implications:

List the potential implications of your project for three of the stakeholders identified. Consider both positive and negative effects, including social, economic, and ethical aspects.

Ethical Considerations:

What goals should we keep in mind to address the broader ethical implications of your project? Think about issues such as privacy, fairness, transparency, bias, and safety.

Examples:

Facial Recognition:

Stakeholders: End-users, law enforcement agencies, minority groups, privacy advocates.

Implications: End-users may benefit from enhanced security but may also face privacy concerns. Law enforcement agencies may improve surveillance capabilities but risk misuse and infringement of civil liberties.

Minority groups may experience disproportionate surveillance and potential discrimination.

Ethical Considerations: Addressing biases in the dataset to prevent discriminatory outcomes. Ensuring transparency and explainability in how facial recognition technology is deployed and used. Safeguarding individuals' privacy rights and providing opt-out mechanisms.

Generative Music Model:

Stakeholders: Musicians, music industry, AI developers, and listeners.

Implications: Musicians may face challenges to their livelihoods if AI-generated music becomes prevalent. In addition, musicians argue that it is unethical to train the model on their original material without consent or compensation.

AI developers may benefit from technological advancements but must consider the ethical use of AI in creative fields.

The prevalence of AI-generated music could undermine the aesthetic and emotional experience for listeners since there will be no emotional connection that an artist establishes with the listener.

Ethical Considerations: Respecting intellectual property rights and ensuring fair compensation for artists.

Encouraging collaboration between AI and human creators rather than replacement.

Addressing concerns about authenticity and emotional impact in AI-generated music.

Ethics Statement Template: (put in your final reports and fill in)

1. Introduction

Student names:

Project Title:

Describe your project, i.e. what does the project do? what does it aim to achieve? (2-3 sentences)

2. Have a large language model (LLM) answer the following questions on your project:

2a. List 3 types of stakeholders that will be affected by the project.

2b. What will an explanation that is given to each stakeholder look like? (1 paragraph max)

2c. Who is responsible for giving the explanation to each stakeholder? (1 paragraph max)

3. Reflect on the AI output (this section requires your independent, creative thinking. Do not use LLMs for this stage!):

3a. What do you think needs to be added/changed in the Generative AI responses, to make the explanation more ethical?

(1 paragraph max)

List of LLMs to use (choose one or more to use while filling in the template):

- ChatGPT: <https://chat.openai.com/>
- Claude: <https://claude.ai/>
- Gemini: <https://gemini.google.com/>
- You: <https://you.com/>
- Mistral: <https://chat.mistral.ai/>